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Sonic Placemaking - Facilitating Listening Behaviour Through Design

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Abstract

In this article, we build on space and place theory to define sonic space and sonic place as fundamental concepts for designers who wish to incorporate audible elements into their work. Based on an interdisciplinary approach that brings together interaction design and sound studies, we propose a Sonic Placemaking Model to guide designers in exploring the sonic environment and address how design choices affect space and place. Based on the development of soundscape designs for three cultural history museums, we unfold the years-long design process and discuss how we applied methods from design and sound studies to understand users as listeners. Lastly, we present three areas of concern when designing for sonic placemaking: 1) going from sounds of objects to sounds of activities, 2) designing for informative and atmospheric experiences, and 3) facilitating spatial and temporal experiences.

Keywords

Space, place, soundscape design, listening, sound, design process, sonic space, sonic place

Introduction

Design processes focus predominantly on visual and tangible aspects, whereas the sonic aspects of a design, such as interface feedback, are often added late in the design process, and are considered elements of an independent job performed by a sound designer.

This article's main contribution is the Sonic Placemaking Model, which aims to guide design teams that work with sound and listening as part of the design process. The model builds on existing space and place theory, including the work of the design researchers Harrison and Dourish (1996), and on

the sonic placemaking theory developed by Lacey (2016b). Our goal is to extend the current space and place theory to include a sonic scope that also focuses on users as listeners.

The article is based on the research-through-design research approach, (Zimmerman, Forlizzi and Evenson 2007), where a team of interaction designer researchers, sound designers, and museum curators worked together to design and implement soundscapes of the past for three cultural history museums to enable museum visitors to experience historical spaces in new ways. The three museums are all placed in the buildings where the history once took place and are briefly presented below.

The Buchholtz House in Struer, Denmark, was the home of Danish writer Johannes Buchholtz and his wife Olga during the 1920s. Their home was described as artsy, warm, and open, as the blue colours of the walls and the odd decor provided the backdrop for the visits of many members of the cultural elite at this time.

The Skanderborg Bunker is situated in the forest of Skanderborg, Denmark, and was the headquarters of the Nazi Air Force in 1944–1945. The bunker was used as a communications bunker, and as such, it was a working space on an everyday basis, although it was used as a shelter only during air raid alarms. In the bunker, visitors find communication devices, such as telephones, Morse code machines, and an Enigma machine.

The farmhouse in Herning is a replica of a typical Danish knitting farm of the 1850s. The farm consists of a large room where the family knitted, spun wool thread, ate, and so on. All these activities centred around a long table, where three generations of the family socialized while working.

At the foundation for extending the current space and place theory to include a sonic scope, the paper starts out by introducing the distinction between space and place, (Harrison and Dourish 1996), and subsequently defines sonic space and sonic place. Moreover, we present sonic placemaking as described by Lacey (2016a; 2016b), focusing on intentions for sonic placemaking. We conclude this section by introducing the Sonic Placemaking Model.

In the second part of this article, we present the process of designing the three soundscapes, which includes methods that may be used to incorporate sonic elements during the design process. The focus is on how the designer may address an intended user experience, but how the design is experienced by the user is beyond the scope of this paper.

The last part of this paper outlines three areas of concern when designing for sonic placemaking, based on a retrospective analysis of the design process, using the Sonic Placemaking Model. Each of the three concerns addresses challenges that emerged during our own design process, and how we met these by applying design and sound study methods and theories.

Space and Place

To address the fact that a sonic environment is an integral part of a complex physical and social environment, we draw on the work of Harrison and Dourish (1996), who have suggested a fundamental distinction between place and space. They define space as the physical, three-dimensional structure and physical organization of our environment, and include objects, tools, and other kinds of artefacts. They define place as our cultural and social understanding of actions and behaviour in space. The relationship between space and place is social, and hence, the same space may facilitate various places; at the same time, place frames human behaviour, for instance, closed doors and privacy signs that are

part of a space, frame activities in the place. The relationship between space and place may be stated as ‘a space is always what it is, but a place is how it’s used’ (Harrison & Dourish 1996, 69).

Harrison and Dourish (1996) are primarily concerned with the design of media spaces, that is, technologies that connect separate physical locations, and argue that ‘the conscious arrangement of elements [of space] that creates a space that accommodates activity’ (Harrison & Dourish 1996, 74).

Sonic Space and Sonic Place

Based on the distinction between space and place, we will introduce the distinction between sonic space and sonic place. As designers, we use the distinction between sonic space and place to study the sonic environment as situated in space: how sounds respond and behave in accordance with the physical space, and how they are then perceived, experienced, and understood as place by those listening. We emphasize that our use of the terms ‘sonic space’ and ‘sonic place’ is situated within a multisensory space and place. Thus, working with sonic space and place involves a framing that zooms in on the sonic aspects of a chosen design location.

Sonic space

Sonic space refers to what we can hear in a space, defined as the existing and the designed and implemented sonic elements in a specific location with a physical layout and physical organization, including humans and material things. The sonic space is where the designer may design for a variety of experiences and desired outcomes by using various approaches, such as adding or subtracting sounds that will alter the existing space. The sonic space may also be transformed, for instance, by adding elements that will alter existing sounds, such as planting hedges in a park to shield it from

traffic noise. The sonic space is the actual and concrete part of the acoustic environment, whereas the sonic place focuses on how the sounds are perceived, experienced, and understood.

Sonic place

Sonic place involves listening behaviour, meaning how we listen in a space. As in Harrison and Dourish's (1996) definition of place, it refers to the way we use and understand space. Thus, sonic place focuses on understanding the ways in which people create meaning through listening in a setting, and how their perception of sounds alters their behaviour in various ways.

Sound studies has developed various methods for exploring a variety of listening behaviours, and therefore provides a strong basis for using an interdisciplinary approach between sound studies and design to study and engage with users as listeners. As part of our design process, we will highlight the three levels of listening attention developed Truax (2001). The highest level, 'listening-in-search', is where meaning is applied to a sensory experience, by intentionally scanning the soundscape for important sounds. The second level, 'listening-in-readiness', refers to a state where the listener is 'ready' to receive important information, but may focus on other things simultaneously, such as looking at cultural artefacts or reading signs. The third level, 'background listening', is where the listener is not actively paying attention to sounds but perceives them unconsciously while in an environment.

Sonic Placemaking

The term placemaking grew out of urban design thinkers William Whyte and Jane Jacobs in the 1960's, who advocated for cities that catered more to people than cars and buildings. In the following decades placemaking is also related to sound, such as in Jan Gehl's famous book "Life between buildings" (1987) in which he described that sound plays a role in urban design and social interaction, and as such

hearing and listening are parts of the urban qualities of life. In another study, Whyte (2001) found that sound in urban settings might cause positive rather than negative associations. What is meant by *placemaking* differs from field to field, but it can be described as creating quality places where people want to live, work, play, and learn (Wyckoff 2014), in which people act to re-imagine their surrounding environment (Strydom et al. 2018).

Spaces are experienced through all the senses. Focusing on sound, our hearing helps us orient ourselves in space, e.g., whether we are in a big or small space based on the echoes that our footsteps produce. As such, sound enhances the spatial dynamics of a place and its meaning (Yildirim et al. 2022).

Sound studies has a long tradition of engaging with the environment through listening, such as describing sound-walking as active listening and engagement with place (Westerkamp 1974). Schafer (1977) developed place-related terms to describe parts of the soundscape; keynote, sound marks, and signal sounds, which each describe cultural and natural sounds that can be found in a space that makes it distinct and recognizable to the people who inhabit the space. Truax (2001) describes how the listener receives environmental knowledge through sound using different levels of attention. LaBelle (2010) describes how sound brings flexibility and uncertainty to the stability of a space, and how acoustic space has no fixed point, but rather is a sphere without fixed boundaries. The acoustic, or sonic space, thus differs from visual understandings of space; we cannot see what is around the corner, but we might be able to hear it.

By approaching sonic space and place as designers, we may say that the act of designing soundscapes is the process of sonic placemaking – creating sonic spaces that allow for various experiences of the space as a sonic place. To define sonic placemaking, we draw on sound studies scholar Jordan Lacey's definition, which builds on Fleming's (2007) concept of placemaking. Lacey extends placemaking to

the sonic elements of the environment, by defining sonic placemaking as localized sonic interventions that interconnect people with the spaces they inhabit (Lacey 2016a).

As part of Lacey's definition of sonic placemaking he introduced ten intentions, which can help guide the process of creating the sonic intervention (Lacey 2016b). The ten intentions were developed for artistic sound installations but may be adapted to other fields. It is our argument that the ten intentions are also beneficial from a design perspective, as they can function as a reflection tool and guidance during the design process. In the beginning, the intentions can help guide a potential design brief; how do we best accomplish our design goals? During the design process, they can help the designer navigate the design space. Next, we will briefly introduce Lacey's ten intentions.

Ten intentions for sonic placemaking

Lacey's ten intentions describe the intended experience that creative practitioners wish their installation to give to its listeners. The ten intentions are useful as part of designing soundscapes, as they may help the designer to communicate and explore: for instance, how various technologies may support a given intention. The ten intentions (2016b, 146–147) are:

1. Reveal: subtract sounds to reveal concealed qualities of a space
2. Captivate: make the listener more attentive to the immediate environment
3. Evoke: draw forth emotive expressions
4. Play: make room for playfulness in everyday spaces
5. Appear: add or accentuate something to make the unnoticed appear
6. Subtract: transform spaces by subtracting noise
7. Quantity: consider the quantifiable aspects of sound: volume, density, duration, etc.
8. Meditate: induce meditative states by adding transformative sounds to existing soundscapes
9. De-alienate: draw alienated everyday listeners back into the moment

10. Engage: make listeners engage in everyday life through interaction

Lacey's (2016b) ten intentions focus on sound and sonic properties that are not included in Harrison and Dourish's distinction between space and place. On the other hand, Harrison and Dourish make a clearer distinction between space and place, which is not defined by Lacey. Thus, we argue that combining Harrison and Dourish's (1996) and Lacey's (2016b) work provides a foundation for designers working with soundscape design. In the following section, we will propose our Sonic Placemaking Model, which brings together the two theories to show how they both relate to sonic placemaking in design.

Sonic Placemaking Model

We propose the Sonic Placemaking Model (Figure 1) based on the distinction between sonic place and sonic space. A sonic space is part of a space and is defined by the natural and the designed sounds. Sonic place focuses on people's listening behaviour as part of their actions and behaviour in a space. From a design perspective, we may design in the space domain to facilitate user experiences, and concrete design decisions in a sonic space may facilitate an intended listening behaviour in a sonic place. For instance, we may transform a physical space by incorporating acoustic panels that absorb sound, or we can install a white noise machine in our bedroom to mask the city noises outside our window. Furthermore, we can facilitate intended listening behaviours through a variety of technologies and transformations of space.

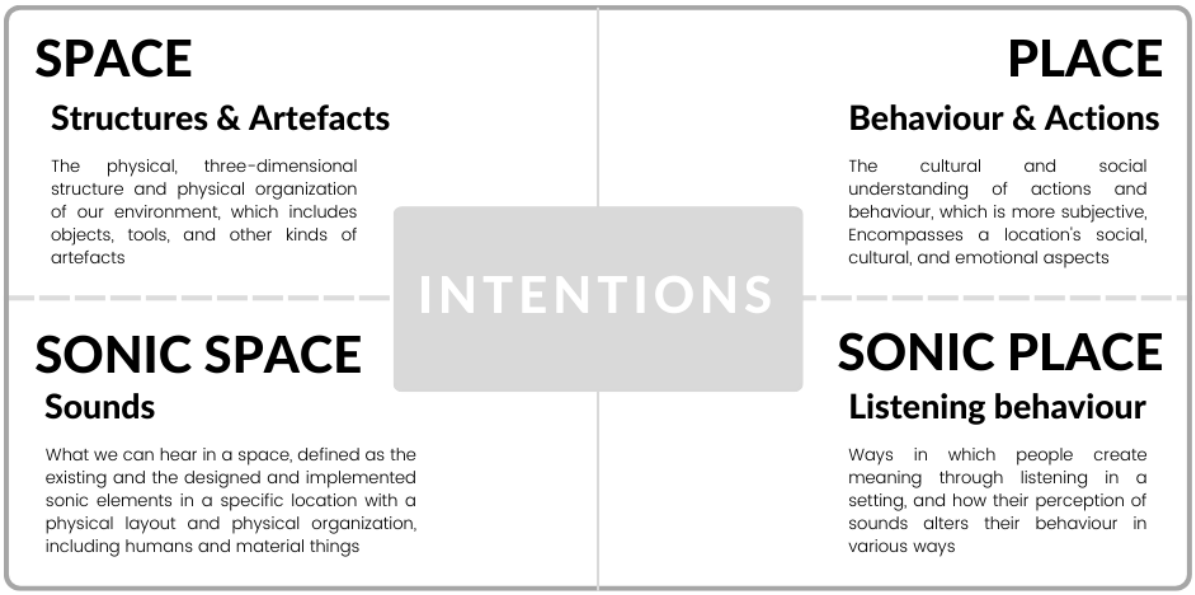


Figure 1: The Sonic Placemaking Model.

The Sonic Placemaking model distinguishes between the physical aspect of space and the sonic aspect of space, and between the listening behaviour and other kinds of behaviours and actions in place, see Figure 1. 'Intentions', see Lacey's ten intentions in section *Ten intentions for sonic placemaking*, operate across space and place. Intentions relate to both space and place, in the sense that an intention is the intended change of place that the designer aims for, which can be enabled by a specific design decision with respect to space. As an example, the intention may be to make the guests stay longer in the exhibition, which can for instance be enabled by designing an engaging installation. Importantly, a designer may *intend* to create a specific user experience, but how the design is experienced, is up to the user.

The model is intended to guide designers who work with soundscape design, by drawing attention to how design choices affect each other. For instance, the designer may transform sonic space by adding materials to physical space, such as planting hedges in a park, or the designer may choose to transform the sonic space by adding or subtracting sounds. Lastly, we may define intentions that focus on a desired listening experience in a sonic place, and from there, determine whether the experience is best realized by working directly in the sonic space or in the broader space.

Next, we present the design process that we implemented to design three soundscapes for historical buildings, followed by a short presentation of the final soundscape designs. We address three areas of concern when designing for sonic placemaking, based on an analysis of the design process and design outcome when using the Sonic Placemaking Model.

The Design Process

The design process began with an analysis of a collection of previous sound design cases. The following presentation of the design process is organized according to the timeline in Figure 2.

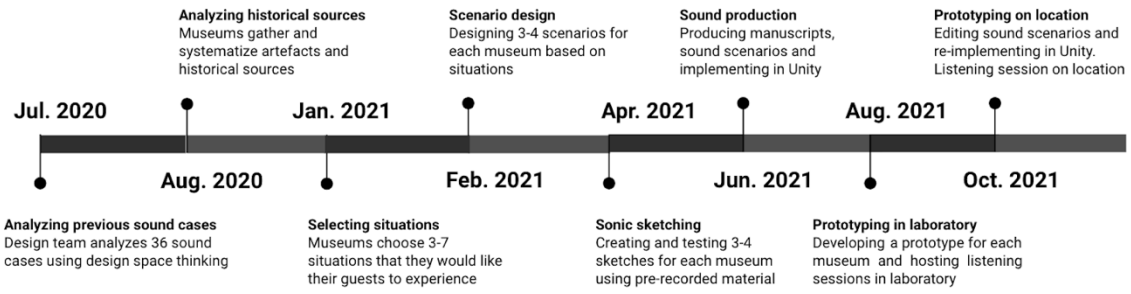


Figure 2: Timeline of design activities (Udsen and Halskov 2022)

Activity 1: Analysing Previous Sound Cases

To create the initial soundscape design space, we followed Halskov et al.'s (2015) three-step process, supported by a Design Space Tool to map the initial design space. First, we selected design cases that used sound installations as part of their environments. Next, we analysed the cases, to identify commonly considered aspects used in the designs. Lastly, the constructed design space was examined to identify densely and sparsely populated areas of the design space. We identified 36 sound installation cases (step 1), which were analysed in order to map the aspects and options reflected by the cases (step 2). The aspects were organized into main categories, for instance INTERACTION, which focuses on technical aspects such as interactivity and mediation, and LISTENING, which

focuses on how listening is tied to the spatiality of the soundscape design. Figure 3 and Figure 4 show the outcome of the analysis as presented in the Design Space Tool, which was used throughout our design process. The aspects addressed in the collections of cases are listed in the second row, and for each aspect the options considered are listed below. The design space may be explored in the tool's search mode, for instance, clicking the small plus (+) next to an option selects only those cases that deploy the option, while clicking the minus (−) signs selects those examples that do not. The design space may be explored through the Sound Installations website.

CONTEXT				AUDIO SYSTEM				
Domain	**Environment**	**Extent**	**Duration**	**Audio source**	**Appearance**	**Transmission channel**	**Spatiality**	**Playback dynamics**
+ − (11) Art	+ − (29) Indoor	+ − (14) Exhibition	+ − (15) Permanent	+ − (10) Headphones	+ − (18) Visible	+ − (6) Mono	+ − (4) 0D	+ − (20) Dynamic
+ − (19) Cultural-historical	+ − (6) Urban	+ − (16) Piece	+ − (19) Temporary	+ − (25) Speaker	+ − (18) Hidden	+ − (7) Stereo	+ − (2) 1D	+ − (8) Looping
+ − (1) Natural history	+ − (3) Nature			+ − (5) Subwoofer		+ − (16) Surround	+ − (8) 2D	+ − (12) Single triggered
+ − (1) Commercial				+ − (1) Transducer			+ − (10) 3D	
+ − (5) Other				+ − (2) Material or object				

Figure 3: Design space tool visualizing the aspects and associated options in the CONTEXT and AUDIO SYSTEM categories. The tool supports the exploration of the design space by selecting (or excluding) combinations of options by clicking small plus (+) and minus (-) signs next to each option. The number of cases sharing each of the options is in parentheses.

INTERACTION		CONTENT		LISTENING		
Interactivity	**Mediator**	**Purpose**	**Types of sound**	**Listening attention**	**Spatiality of listening spaces**	**Intention**
+ − (2) Interactive	+ − (0) Touch screen	+ − (2) Informative	+ − (3) Soundscape	+ − (0) Listening-in-search	+ − ⓘ (4) Public	+ − ⓘ (1) Captivate
+ − (1) Non-interactive	+ − (1) Mobile devices	+ − (3) Atmospheric	+ − (2) Music	+ − (0) Listening-in-readiness	+ − ⓘ (0) Semi-public	+ − ⓘ (4) Appear
+ − (1) Reactive	+ − (0) Tangible	+ − (1) Animation	+ − (1) Talk	+ − (0) Background listening	+ − ⓘ (0) Personal	+ − ⓘ (0) Subtract
	+ − (2) Sensors		+ − (2) Sound effect			+ − ⓘ (1) Engage/Play
	+ − (0) Bodily					

Figure 4: Design space tool visualising the aspects and associated options in the INTERACTION, CONTENT, and LISTENING categories. Figure legend, see Figure 3

Some of the aspects (e.g. interactivity) originate in interaction design, others are fundamental acoustic aspects (e.g. transmission channel). Moreover, sound studies theory supports the LISTENING category, whose 'Listening attention' aspect is based on Barry Truax's (2001) three levels of listening attention, whereas 'Intention' is based on Lacey's (2016b) intentions, from the sonic rupture model.

Activity 2: Analysing Historical Sources

The next activity conducted by the museums focused on investigating the places' past, that is, how historical figures once inhabited the buildings. The historians gathered and systematized artefacts and historical sources, including letters and journals. The information about the historical objects was organized in Excel sheets, and their sonic significance was rated.

Activity 3: Identifying Situations

Based on the analysis of the historical sources, the historians subsequently identified and described specific situations of sonic significance from historical sources where the historical figures interacted with objects. This activity was supported by a situation questionnaire developed by the design team (See Table 1).

Table 1: Situation Questionnaire

In which room(s) is the situation unfolding?
Which historical figures are present?
Which objects are included? What do they sound like? Do they have several sounds? Are other types of sounds present (e.g., natural, environmental)?
What is the atmosphere/mood of the setting? Do you have examples from historical sources?
Why is this situation interesting? What does it convey, and how does it support the overall narrative structure?
With regard to the guests, what is the most important thing to convey to them through this situation?
Is there a specific object that should be emphasized in this situation?

Activity 4: Scenario Design

Based on the situations mentioned above, the design team created a set of scenarios for each museum (Carroll 1999). For example, Figure 5 below presents two scenarios from the Skanderborg Bunker.

Scenario 1:
The guest walks into the command room and look at artifacts on display. The guest begins to perceive that something is going on in the other rooms; someone is talking to each other, there is typing on a typewriter. Suddenly, a telephone rings in the communication room and the guest can hear a soldier running into the room and answering the phone. The guest can feel tension is building; soldiers are working in a high-pressure situation. the guest can hear a deep humming sound from enemy planes that are approaching; a soldier yells "They are coming!".

Scenario 2:
At the entrance the guest is handed a set of headphones and a little device that can be used to activate stories throughout the exhibition. To play a story the guest has to look for places with small speaker-icons. the guest walks in the weaponry and sees a speaker-icon near one of the weapons. The guest scans the icon with their device and a story begins. It is about two soldiers who speak about being stationed at the East front; rumors of the conditions there, the cold, tales from wounded soldiers.

Figure 5: Examples of scenarios at the Skanderborg Bunker

Activity 5: Sonic sketching

As a next step, the scenarios were explored through sonic sketching (Nykänen et al. 2015; Delle Monache & Rocchesso 2019), similar to visual sketching in interaction design (Buxton 2010), with the main difference that sound is ephemeral, as it may be heard only as long as it is produced, whereas a visual sketch is persistent (Dix & Gongora 2011).

For each museum, a set of two- to four-minute-long sonic sketches was created, using pre-recorded sounds from online sound libraries, and recordings of team-members acting from a script.

Activity 6: Sound Production

The first step of sound production was script-development. The scripts included dialogue and directions for the sound designer concerning sounds of important objects and the atmosphere. We also included short descriptions of the way the historical figures were moving around, for instance, 'soldier runs from communication room to commando room'. Based on the scripts and the knowledge gained from experiencing the sonic sketches on location, the sound designer produced the sounds that were to be part of the final installation. The sound scenarios included four main kinds of sounds: 1) recordings of museum artefacts in action, such as recordings of using a typewriter, 2) recorded

dialogues between voice actors, based on scripts, 3) on-site recordings of the physical space, such as creaking wooden floors, and 4) foley sound effects.

Recording on-site was particularly valuable – for instance recordings of authentic footwear on a building's wooden floor – as this allowed the sound design to mimic the creaking sounds (which were already a part of the guest experience when walking about), and to use these sounds to create the illusion of the historical figures walking among the guests.

Activity 7: Prototyping in the Laboratory

The next step was to incorporate the audio material into a 3D model of each of the buildings, using SoundOmatic, our lab's custom-developed soundscape design tool built on Unity. The sounds are now seen as spatial clusters connected to the corresponding speakers, and coded to act in dynamic ways (intervals, randomization, volume, etc.). To test the prototypes, we invited the museum staff to a listening session in our laboratory, where 14 speakers were positioned in a 3D array in a large room where the location of each speaker corresponded to a zone or a room in the historical buildings, based on their floor plans (e.g., floor plan of Buchholtz House, Figure 6).

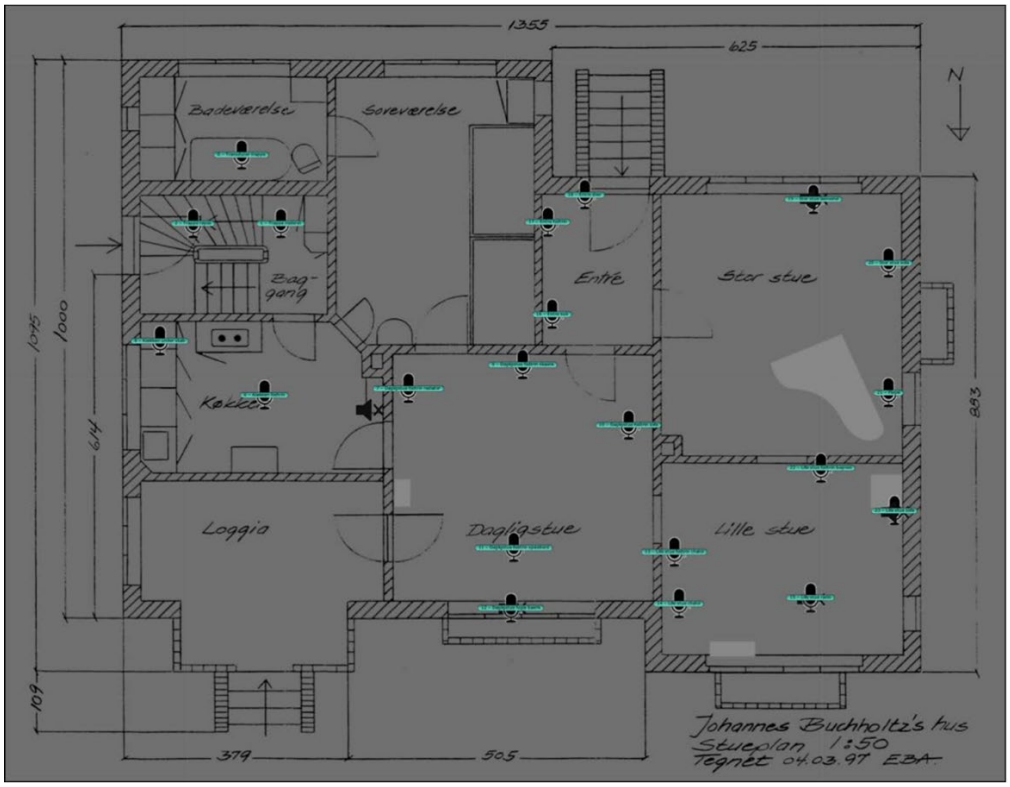


Figure 6: Buchholtz House: Floor plan marked with positions of speakers

The museum staff were asked to give feedback concerning the overall atmosphere and dialogues. When listening to the soundscape design in the laboratory, rather than in the actual buildings, the scale sometimes sounded off; for example, the footsteps of soldiers moving between rooms sounded strange, as the distances between the speakers were not identical to the measurements of the actual spaces of the historical buildings.

Activity 8: Prototyping on Location

Based on the insights gathered from prototyping in the laboratory, we created a full-scale prototype, in which the sound designer corrected the sound scenarios used in the first prototype, before the sounds were deployed in the final 3D Unity model.

An important aspect of prototyping on location was the adjustment of the soundscape design in relation to the spatial aspects of the building, which was primarily done in two ways: first, by hiding the

speakers in the physical space, for instance, one speaker which plays tunes from a radio was built into the actual radio in the room, whereas others were hidden among coats, under couches, and so on; secondly, by adjusting the sound scenarios to the physical space. Thus, prototyping on location helped us ensure that the sounds were adjusted in response to the spatial aspects of the historical buildings.

The Three Soundscape Designs

The main goal of each sound design was to recreate a soundscape of the past, and allow museum guests to step into the everyday life of historical figures. Thus, the guests can listen in on everyday dialogues between historical figures, doing their everyday chores, and so on. We intended to recreate the subtle soundscapes that would have been part of each building, in the time periods in question.

The final soundscape designs consist of a setup of 30 speakers hidden throughout each building. Each soundscape consists of more than one hundred short tracks that are coded to play in certain patterns over a 20- to 25-minute period. Each soundscape follows a structure of three basic levels; background (sounds without human presence), micro-narrative (interactions between human and object) and scenarios (dialogues and interactions between people), see examples in Figure 7. The intervals of a sequence are determined based on these levels, for instance, a background sound may be present constantly, whereas a scenario that includes a dialogue will be played only once within a 20- to 25-minutes period, to ensure that a guest does not hear it twice.

	Buchholtz House	The Bunker	The farmhouse
Sound scenarios	• Writing on a book in the office • Playing the piano and singing with guests • Reading aloud a letter	• The overflight • Soldiers having everyday conversations • Strategies discussed around the map table	• A salesman comes by • A mother soothes a crying baby with a song • Folk tale told around the long table
Micro narratives	• Johannes walks upstairs to work • The radio is turned on • Olga hums in the kitchen	• The telephone rings and a soldier answers • Decoding a message on the Enigma machine	• Sleeping/snoring in the • alcove • A woman sweeping the floor Making thread on a spinning wheel while humming
Background	• Grandfather clock • Buzzing fly • Birds chirping	• Air ventilation • Generator buzzing • Communication devices beeping	• Birds chirping • Firewood in a furnace • Farm animals

Figure 7: Overview of the three levels of soundscape design, with examples

Three Areas of Concern

In this final part of our article, we wish to highlight three areas of concern associated with designing for sonic placemaking. The three areas of concern were identified retrospectively, after the three soundscape designs were developed, and exemplify how the Sonic Placemaking model may help designers when they design for sonic placemaking; this includes how Lacey's (2016b) intentions may support sonic placemaking

From sounds of objects to sounds of activities

The first area of concern to which we refer is *going from sounds of objects to sounds of activities* came to our attention at the beginning of the design process, mainly during design activities 2 and 3.

We regard two of Lacey's intentions as central to this area of concern: *appear* and *captivate*. *Appear* refers to the process of adding or accentuating something to make the unnoticed appear. From the beginning of this project, the goal was to add sounds to the buildings in order to bring back to life the

everyday life of historical figures. Furthermore, the focus on making the unnoticed appear matches the three museums' desire to focus on specific situations through a combination of the objects in the building and storytelling. Therefore, we also include the intention *captivate*, that focuses on making the listener more attentive to the immediate environment. At the beginning of the design process this intention also suggested how we might capture knowledge and stories from historical sources, so these could be transformed into sound. Thus, the intention to *captivate* is twofold, as it refers to both captivating a listener in a sonic place and requires the designer to think about how the museum's dissemination of the history in space may be supported by sounds in a sonic space.

During the analysis of the historical source material, museum staff pointed out that there were not that many objects in the buildings that 'had a sound', therefore they were not capturing the listeners attention and able to make them attentive to the environment. For instance in the Buchholtz House, apart from the clock, the piano, and the radio, not that many objects actually made a sound. As part of design activity 2 the museum staff made detailed lists of all the objects in their buildings and rated them with regard to what they called their 'sonic significance'. As many of the objects did not 'have a sound', they were deemed irrelevant to this context. So, we needed another way to make the 'sonic significance' appear, and from this, to explore how we could determine which objects were part of the everyday life of the historical figures, so that we could base our sound design on these interactions. Thus, this area of concern focuses on 1) how to capture and systematize the objects that the museums wish to highlight in their buildings, and 2) how to understand the activities in which the historical figures used the objects, and how to make them appear through sound.

Designing for informative and atmospheric experiences

The next area of concern was mainly addressed through activities 4 (Scenario Design) and 5 (Sonic Sketching) but was also addressed by activity 1 (Analysing Previous Sound Cases) through the Content

aspect (see Figure 4). With respect to this area of concern, we began to define the intended listening experiences (see the Listening aspect, Figure 4) that the installations should support. Four intentions are relevant to informative and atmospheric experiences: *appear*, *captivate*, *evoke*, and *quantity*.

The first intention, *appear*, is used to focus on parts of the museums' dissemination of history that are not currently perceivable through objects. As we have noted, the stories of the everyday life of the buildings' past are coming to life through sonifications of the interactions between people and objects. For instance, in the small living room of Buchholtz House there is a desk and a sofa, which, as objects, do not tell the guests much, but we need to reveal how they were used by historical figures in this place. To address this intention, we developed scenarios for each museum, to begin to explore not only how we could make the unnoticed noticed, but also how sounds could be added to the spaces.

The intention, *captivate,* became central with respect to designing for informative and atmospheric experiences, as we intended to draw attention to the building through the soundscape design. As part of the design process, we found it useful to define the kind of attention we intended to attract from the guests. Early in the process, the museums stated that they wanted their guests to be able to immerse themselves in the sound installations as a new way of experiencing the museum, but that the soundscape design should also allow guests to experience the exhibition without being distracted by the sounds. So, the soundscape design should both allow for immersive and more subtle ways of listening. To explore this, we applied Barry Truax's (2001) three levels of listening attention: listening-in-search, listening-in-readiness, and background listening. The three levels of attention enabled us to explore the various ways a guest might experience a sonic place. A guest who was listening-in-search or -in-readiness would find that the dialogues that are part of the soundscape design offer a more informative experience. For example, in the farmhouse, guests can hear a scenario where a dealer from town who comes to the farm to buy knitted woollen clothes. If the guest listens closely to the dialogue

between the farmer and dealer, they will be able to gather information about the types of knitted clothing the family produced, prices, where the clothes were sold at markets in nearby towns, and so on. If the guest hears the scenario with background listening attention, they may only register that someone is coming to the farm and having an ordinary conversation. Therefore, a guest who focuses more on the visual experience (reading signs, looking at objects) of the space will perceive the sounds less consciously, and thus have a more subtle and atmospheric experience.

The third intention, *evoke*, is closely related to guests' subtle atmospheric experiences through background listening. *Evoke* is tied to the soundscape design's ability to elicit emotive expressions. The kinds of emotive expressions that we intended in the three installations are very different, as their histories are very different. For instance, the Buchholtz Home is meant to have a warm, artistic, eccentric feeling, whereas the Skanderborg Bunker is meant to convey a more stressful, strict, sometimes bored atmosphere. How this intention is expressed may be observed as part of sonic place. For instance, following the implementation, we have observed that in Buchholtz House, the older audience (70+) enjoys the nostalgia evoked by the sounds of the old grandfather clock and the radio, which remind them of childhood visits at their grandparents' homes. Another very different emotional response may be observed in the Skanderborg Bunker, where an air raid siren goes off in the building as part of a scenario. Here, the museum staff found that they need to warn parents with small children, who may be frightened by the sudden loud noise. Lastly, in the Farmhouse we added sleeping sounds in the two alcoves, and we have observed that children laugh at the snoring, and that it makes them investigate the alcove to see whether they can locate the snorer.

The last intention, *quantity*, captures the designer's decisions regarding the sound's volume, density, duration and so on, which was addressed by activity 5 (Sound Production). For this area of concern, *quantity* is especially important regarding volume, as this allows us to draw attention to certain parts

of the soundscape design as either informative or atmospheric. For instance, we could increase the volume of a dialogue, to ensure that people can hear it, and we may use it to draw attention (listening-in-readiness) to other rooms, thereby making the guests move from one room to the next. At the same time, volume contributes to the illusion of being back in time, as part of the historical figures' everyday life, and therefore, if the volume is too high, we risk ruining the illusion, and draw adverse attention to a part of the soundscape.

Facilitating spatial and temporal experiences

The last area of concern was mainly addressed by the last two design activities, 7 (Prototyping in Laboratory) and 8 (Prototyping on Location), in which we focused on how to bring together all four elements of the Sonic Placemaking Model, Figure 1, space, place, sonic space, and sonic place. To create soundscape designs that are site-specific and made for actual buildings, based on the sounds of everyday life, is a way to allow guests to have the feeling of stepping into the past, into the everyday life of historical figures. To achieve this, we used our research laboratory's custom-developed software, called Sound-O-Matic, which was developed to create 3D soundscapes that are coded into a 3D model of each building, which is then implemented in the actual building. Instead of a linear track that uses playback, the composition consists of many short tracks that convey scenes, such as 'a soldier walks from A to B' or 'a telephone rings and soldiers pick it up'. Furthermore, the short tracks are coded so that the same historical figure cannot be in two places at the same time, and more prominent scenes are coded so that they occur only once in a 20- to 25-minute period, so that the typical guest will not hear the same scene twice.

With regard to Lacey's intentions, we regard three as central to this area of concern: *appear*, *captivate* and *quantity*. For instance, *appear* was relevant to design activity 8 (Prototyping on Location) in the Buchholtz House, where the design team and museum decided that sounds should be added that could

help to emphasize the first floor of the building, so as the guests would not overlook it. Therefore, we decided that one of the scenes should include a dialogue between Olga (wife) and a postman, and she tells him that Johannes Buchholtz (the author) is upstairs, working on a new book. Furthermore, we decided to add a transducer under the stairs between the two floors, which mimicked the creaking of floorboards when going up or down the stairs. Thus, we could create scenes where historical figures were walking up the staircase, so that the guests could hear the creaking floorboards from the ground floor, thus making them attentive to the fact that something was happening upstairs.

The next intention, *captivate*, also concerns the staircase. The intention to *captivate* is to make the listener more attentive to their environment, which, for this area of concern, we refer to as making them attentive to the sub-spaces in the building. As museum guests, we are used to being guided by signs that tell us which room to enter next, but in these historical buildings, the guests have to explore and find their own way. Therefore, guests may not pay attention to a room, as they are simply unable to see it. Here, sound becomes a powerful tool, as it can travel through walls, making us notice that there is something on the other side.

The last intention, *quantity*, was especially important to facilitate spatial and temporal experiences, as we here focused on adjusting the soundscape design to the physical space of the building. Thus, we had to hear how the sonic space fitted the broader space, as the physical layout has a great influence on how sounds behave within it. For instance, the Skanderborg Bunker is made of concrete, which makes the sounds bounce off the walls, so we had to add a background air ventilation sound to hide some of that, whereas the Farmhouse has open doors and windows, which let in traffic noise from outside, which we had to cover with sounds from nature, such as birds, and farm animals. To adjust the soundscape designs to fit each building, we held an on-site listening workshop (Design activity 8), where the design team and museum staff listened to parts of the soundscape design while standing in

various rooms, to hear how each sounded from various positions, as well as how they sounded if we had a high level of listening attention, and how it sounded if we paid less attention while talking to a colleague and walking about.

Conclusion

The main contribution of this paper is the Sonic Placemaking model that may support designing for sonic placemaking. The model extends the original space-place model by addressing the sonic aspect in addition to the spatial and physical aspects of a space. The Sonic Placemaking model also highlights listening behaviour as part of people's behaviour in place. Moreover, we have argued that an awareness of design intentions may support the process of sonic placemaking. Last, but not least, we have suggested three areas of concern when designing for sonic placemaking: 1) going from sounds of objects to sounds of activities, 2) designing for informative and atmospheric experiences, and 3) facilitating spatial and temporal experiences.

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